



School of Pure and Applied Sciences

FEDERATED LEARNING FOR MEDICAL SIMULATION

Submitted by

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DECLARATION

We declare that this work has not been submitted in this or any other University for the awarding of the course marks. To the best of our knowledge and belief, this work contains no material previously published or written by another person except where due reference is made.

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DECLARATION

I, Gilbert Langat, confirm that I have reviewed the work presented in this proposal and that it meets the required academic standards. I have provided guidance and supervision throughout the project, and I am satisfied with the originality and integrity of the work submitted.

Supervisor: Gilbert Langat

Signature: Date:

Dedication

We dedicate this project to the divine guidance of God, and to our families and friends whose consistent encouragement has been a constant source of strength. Special thanks are extended to our classmates for their diligent efforts, steadfast dedication, and unwavering support. We express gratitude to the instructor who generously dedicated time to guide us. This dedication is a token of appreciation to all those who have contributed to the success of this project, making it a collaborative effort and a reflection of shared dedication and support.

Acknowledgement

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Abstract

Federated Learning (FL) has emerged as a pivotal distributed machine learning approach that enables multiple clients to collaboratively train a model without sharing their local data (Kairouz et al., 2021). This paradigm is particularly transformative for the healthcare sector, where data privacy is governed by strict regulations like HIPAA and GDPR, and centralizing sensitive patient data, such as medical images, poses significant security and privacy risks (Rieke et al., 2020). The current state-of-the-art in medical FL demonstrates feasibility but reveals critical gaps. Standard FL algorithms, like Federated Averaging (FedAvg), are susceptible to privacy inference attacks and often experience significant performance degradation when faced with non-Independent and Identically Distributed (non-IID) data, a common characteristic of medical datasets across different hospitals (Li et al., 2020). While Differential Privacy (DP) has been proposed as a solution to FL's privacy vulnerabilities, its application in medical imaging lacks a systematic analysis of the privacy-utility trade-off under realistic, heterogeneous data conditions (Wei et al., 2020). The motivation for this project stems from the urgent need to develop a robust, privacy-preserving FL framework that can function effectively in real-world clinical environments where data is inherently distributed and non-IID. This research is driven by the goal of enabling collaborative AI advancement in medicine without compromising patient confidentiality. The preliminary outcome of this project is the development and open-source release of ARCH-FL, a software framework that integrates DP with FL specifically for medical image analysis. The expected result is a comprehensive empirical evaluation quantifying how different levels of DP noise impact model accuracy, convergence, and fairness across various non-IID data partitions. The tools to be used include the PyTorch deep learning library, the Opacus library for implementing DP-SGD, the MedMNIST and COVIDx CXR datasets for medical image classification, and MLflow for experiment tracking. This work aims to provide practical guidelines and a validated baseline for deploying privacy-preserving federated learning in healthcare.

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Abbreviations

Abbreviation	Full Form
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FL	Federated Learning
DP	Differential Privacy
ARCH-FL	Architecture for Federated Learning
IID	Independent and Identically Distributed
non-IID	Non-Independent and Identically Distributed
DP-SGD	Differentially Private Stochastic Gradient Descent
ϵ (epsilon)	Privacy Budget
δ (delta)	Privacy Failure Probability
CXR	Chest X-Ray
CNN	Convolutional Neural Network
FedAvg	Federated Averaging
ML	Machine Learning
AI	Artificial Intelligence
EHR	Electronic Health Records
HIPAA	Health Insurance Portability and Accountability Act
GDPR	General Data Protection Regulation

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Chapter 1: Introduction

1.1 Introduction to the Chapter

This chapter establishes the foundational context for the research project, ARCH-FL. It begins by outlining the background of federated learning and its significance in the medical domain, followed by a critique of the current systems and a clear articulation of the existing problems. The chapter then presents the proposed system as a solution, detailing its purpose, objectives, and justifications. Finally, it defines the scope, acknowledges limitations, and clarifies the operational terms used throughout the study.

1.2 Background

The proliferation of Artificial Intelligence (AI) in healthcare, particularly in medical image analysis, has demonstrated remarkable potential for improving diagnostic accuracy and efficiency (Esteva et al., 2019). However, the development of robust AI models requires access to large, diverse datasets, which are often siloed across different hospitals due to patient privacy concerns, proprietary interests, and stringent data protection regulations (Kaissis et al., 2020). This data fragmentation creates a significant bottleneck for medical AI research. Federated Learning (FL) was introduced by McMahan et al. (2017) as a decentralized machine learning approach that circumvents the need for data centralization. In a typical FL system, a global model is trained by aggregating model updates learned from local data on client devices or institutions, while the raw data never leaves its original location. Despite its promise, vanilla FL is not inherently private; model updates can leak information about the underlying training data, making them vulnerable to membership inference and reconstruction attacks (Nasr, Shokri, & Houmansadr, 2019).

1.3 Current/Existing System

The current state-of-the-art for multi-institutional medical AI research often involves cumbersome data sharing agreements and the creation of centralized data repositories. Where FL is implemented, it frequently relies on the basic Federated Averaging (FedAvg) algorithm without formal privacy guarantees. Existing open-source FL frameworks (e.g., TensorFlow Federated, PySyft) provide the architectural backbone but offer limited, non-standardized integrations with rigorous privacy-preserving

technologies like Differential Privacy (DP) for medical imaging tasks. Furthermore, most research and implementations assume IID data distribution across clients, an assumption that is routinely violated in real-world healthcare scenarios where patient demographics and disease prevalence vary significantly between hospitals (Sheller et al., 2020). This gap between research assumptions and practical realities hinders the effective deployment of FL in clinical settings.

1.4 Problem Statement

The core problem is the absence of a systematically evaluated, open-source framework that integrates Differential Privacy with Federated Learning specifically tailored for the challenges of medical image analysis, particularly under non-IID data distributions. This leads to a critical dilemma: how can hospitals collaboratively train accurate AI models without sharing data and without sacrificing patient privacy or model utility?

The specific issues addressed are:

- 1) The vulnerability of standard FL to privacy attacks in a medical context.
- 2) The performance degradation of FL models under realistic non-IID medical data.
- 3) The lack of clear guidelines on the trade-off between privacy budget (ϵ) and model utility (e.g., accuracy) in federated medical imaging.

1.5 Proposed System

This research proposes ARCH-FL: a novel, open-source framework for Federated Learning with Differential Privacy designed specifically for medical image analysis. The system implements a secure coordinator server that aggregates locally trained model updates from multiple simulated hospital clients. Each client employs DP-SGD during local training to ensure that the model updates provide formal (ϵ, δ) -differential privacy guarantees. The framework is designed to systematically test and evaluate model performance under various levels of privacy protection and different types of non-IID data splits, providing a comprehensive analysis of the privacy-utility trade-off.

1.6 Purpose of the Study

The purpose of this study is to design, implement, and empirically evaluate a differentially private federated learning framework that enables effective and secure collaborative training of medical image analysis models across distributed data sources.

1.7 General Objective

To develop and validate the ARCH-FL framework for privacy-preserving federated learning in medical image analysis.

1.8 Specific Objectives

1. **To design** a system architecture that integrates differential privacy mechanisms within a federated learning pipeline for managing distributed medical image datasets.
2. **To implement** the ARCH-FL framework using PyTorch and Opacus, featuring configurable privacy parameters (ϵ , δ) and non-IID data partitioning strategies.
3. **To conduct** a series of experiments using medical image datasets (e.g., PneumoniaMNIST) to establish baseline performance without privacy and under IID conditions.
4. **To analyze** the impact of varying differential privacy budgets and non-IID data distributions on model utility, measured by classification accuracy, convergence speed, and fairness across clients.
5. **To disseminate** the research findings and the open-source codebase to the community, providing practical guidelines for the implementation of private FL in healthcare.

1.9 Justification

This research is justified by the critical need to advance medical AI in an ethical and legally compliant manner. It addresses the direct concerns of healthcare institutions regarding data privacy and security. By providing a validated framework and concrete evidence on the privacy-utility trade-off, this study lowers the barrier for hospitals to engage in collaborative research. This may potentially accelerate the development of more robust and generalizable diagnostic AI models without compromising patient trust or violating data protection laws.

1.10 Scope

The scope of this study is confined to image classification tasks using 2D medical images, primarily benchmarked on the MedMNIST collection (specifically PneumoniaMNIST) and potentially validated on a subset of the COVIDx CXR dataset. The federated learning environment will be simulated on a single machine, with clients represented as separate data partitions. The research will focus on the FedAvg aggregation algorithm and the DP-SGD mechanism for privacy.

1.11 Limitations

The primary limitation of this work is the use of simulated clients and publicly available datasets, which may not fully capture the complexities of a real-world multi-hospital network, including communication overheads, systems heterogeneity, and adversarial client behavior. Furthermore, the study is limited to classification tasks and may not generalize to more complex medical imaging problems like segmentation or detection without further modification.

1.12 Significance of the Study

This study is significant because it provides a practical, open-source tool and a comprehensive empirical analysis that bridges a critical gap in privacy-preserving medical AI. The findings will offer much-needed insights and benchmarks for researchers and practitioners aiming to deploy FL in clinical settings, thereby contributing to the responsible advancement of AI in healthcare.

1.13 Operational Definition of Terms

- **Federated Learning (FL):** A distributed machine learning technique where a model is trained across multiple decentralized devices or servers holding local data samples, without exchanging them.
- **Differential Privacy (DP):** A rigorous mathematical framework for ensuring that the output of a computation does not reveal information about any individual data point in the input dataset.
- **Privacy Budget (ϵ):** A parameter that quantifies the privacy loss; a smaller ϵ implies stronger privacy guarantees.

- **Non-IID Data:** Data that is not Independent and Identically Distributed across clients. In this context, it refers to uneven distributions of disease labels or image types across different hospital datasets.
- **Model Utility:** The performance of the machine learning model, typically measured by metrics such as **accuracy**, **precision**, **recall**, and **F1-score** on a held-out test set.
- **DP-SGD:** Differentially Private Stochastic Gradient Descent, an optimization algorithm that adds calibrated noise to gradients during training to achieve differential privacy.

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